

News Framing Analysis using Mainstream Media

Context Analysis using The New York Times and The Guardian

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Abstract—This research explored media coverage of the pandemic in the United States across the Trump administration. The study revealed that both The New York Times (NYT) and The Guardian discussed the pandemic and utilized the two listed framing attributes, but the salience of these attributes varied for each media outlet. Key findings from examining individual sources showed the nuances of the media’s coverage and the differences in framing the pandemic across the both mainstream.

I. INTRODUCTION

Framing theory is formally defined by Entman (1993) [1] as the formulation, designation, and presentation of aspects of a topic that “promote a particular problem, causal interpretation, moral evaluation, and/or treatment recommendation.” The effect of framing in media analysis plays a crucial role in shaping public perceptions and discourse. A study by Dee K. et al [2] focused on the differences in attribute salience and partisan framing between liberal-leaning and conservative-leaning media streams identified some of the key attributes: “economic consequences” and “fear/scaremongering.” The findings of this study indicate that conservative-leaning media tend to employ more partisan language and express a greater degree of negative sentiment towards preventive measures. In contrast, liberal-leaning media are more inclined to emphasize prevention and protection strategies, report on COVID-19 statistics, and convey a more positive sentiment regarding measures aimed at public safety. This divergence in framing not only highlights the varying narrative approaches of different media outlets but also underscores the significant impact of these frames on audience perception and behavior.

The linguistic framing approach has garnered increased attention within the Natural Language Processing community, particularly as advanced computational methods are employed to process, analyze, and label texts and speech across various scholarly domains, including international conflicts, sustainability, and beyond.

The analysis examines how two prominent English-language mainstream news outlets, The New York Times (United States) and The Guardian (United Kingdom), framed Trump’s COVID-19 response during the final year of his administration from January 2020 to January 2021. The previous studies validate this analysis, as it explores a two-dimensional lexicon-based framing approach to capture specific dimensions of media framing that directly operationalize responsibility attribution and science engagement.

II. MATERIALS AND METHODS

A. Time Frame

The data for this investigation was collected over the period from January 20, 2020, to January 19, 2021. This time-frame encompasses the entirety of the Trump administration’s response to the COVID-19 pandemic, beginning with the first reported case of the coronavirus in Snohomish County, Washington, on January 20, 2020 [3], and concluding shortly before the administration ended on January 19, 2021 [4].

The central research questions guiding this analysis are as follows:

1. **RQ1 (Responsibility Frame):** How do The New York Times and The Guardian differ in their attribution of responsibility for COVID-19 outcomes during the Trump administration? Specifically, how do these outlets delineate blame for pandemic-related failures versus acknowledgment of successes?

2. **RQ2 (Science Frame):** In what ways do these two publications frame Trump’s engagement with scientific expertise throughout the pandemic?

B. Data Collection and Preprocessing Methods

The articles were gathered through the respective outlet APIs, employing a pre-filtering strategy during the scraping process to focus specifically on content related to COVID-19 and former President Trump. The search queries encompassed the following terms, “Trump”, “coronavirus”, “COVID-19”, and “pandemic”.

These terms were employed in conjunction with appropriate boolean operators to ensure the relevance of the collected articles. This pre-filtering methodology guaranteed that the corpus exclusively included pandemic-related coverage referencing these selected queries.

Following the data collection phase, cleaning procedures were implemented, which involved standardizing data formats, merging headlines with abstract text fields, eliminating duplicate articles, and filtering for specific time ranges. The final dataset comprises 8,807 articles, with 180 sourced from The New York Times and 8,627 from the Guardian. This significant disparity in article counts (a ratio of 48:1) highlights both a genuine difference in publication volume and a technical limitation posed by the NY Times API, which restricts pagination to approximately 1,000 results per query. This imbalance is an important limitation that may affect the statistical power of any cross-outlet comparisons. Nevertheless, both samples

Table I
RESPONSIBILITY FRAME LEXICONS

Positive Lexicon (50 terms)	Negative Lexicon (88 terms)
Leadership & Achievement Terms	
leadership, led the response, took charge, decisive, decisive action, bold action, swift response, quick action, acted quickly, immediate action, strong leadership, took responsibility, stepped up, rose to the challenge, credited with, praised for, effective response, successful, accomplishment, delivered, achievement, managed well, under control, got it under control, controlled the outbreak, handled effectively, competent response, saved lives	failed, failure, failed to act, failed to respond, negligent, negligence, mismanaged, mismanagement, bungled, botched, fumbled, disaster, fiasco, incompetent, incompetence, reckless, dereliction
Coordination & Strategy	
coordinated response, national strategy, federal coordination, mobilized, organized effort, task force, unified response, working together, coordinated effort, effective coordination	no plan, no national plan, lack of plan, no strategy, no national strategy, chaotic, chaos, disorganized, unprepared, unpreparedness, caught unprepared, abdicated, abdicated responsibility, abdication, lack of leadership, leadership vacuum, absence of leadership
Warning Response & Timing	
	ignored warnings, dismissed warnings, early warnings ignored, downplayed, downplayed the threat, minimized, minimized the risk, delayed response, slow response, late response, slow to act, months of delay, crucial weeks, lost time, wasted time, squandered time, precious weeks, critical window
Vaccine & Medical Achievement	
operation warp speed, warp speed, vaccine development, vaccine delivery, rapid vaccine, delivered vaccines, vaccine success, record time, unprecedented speed, historic achievement, fastest vaccine, vaccine miracle	
Accountability & Impact	
	responsible for deaths, preventable deaths, avoidable deaths, unnecessary deaths, blood on his hands, could have prevented, should have acted, could have saved, if he had acted, lives lost, needless deaths, tragic failure, catastrophic, catastrophic failure, crisis mismanagement, toll, death toll, rising deaths, mounting deaths, casualties, hundreds of thousands dead, mass death, surging cases, spiraling, out of control, to blame, blamed for, fault, at fault, culpable, accountability, held accountable, must answer for, bears responsibility, share of blame, falls on

are sufficiently large to permit meaningful lexicon-framing analysis.

The combined corpus contains a total of 89,439 sentences, integrating headlines and abstracts from both sources. For The New York Times, the text field comprises headlines and abstracts, whereas for the Guardian, it includes headlines and the snippet field provided by the API. This methodology prioritizes the most prominent content that influences the framing of the articles.

The temporal distribution analysis reveals discernible patterns in coverage intensity, with both outlets exhibiting heightened reporting during the March-May 2020 period (initial pandemic surge), the July-August 2020 timeframe (summer outbreak surge), and October-November 2020 (election period and Trump's COVID-19 diagnosis). The Guardian demonstrated more consistent month-to-month variability in coverage, while The New York Times exhibited a greater degree of variability, likely reflecting divergent editorial priorities and resource allocation strategies.

The lexicon-based framing detection framework centers on two custom-built lexicon sets developed through an interactive process. The framework incorporates valence code analysis

(Bennett, W. L., 1990)[5] for concurrent and full network analysis, assigning codes of plus sign (+) when voices supported administration contra policy, negative sign (-) when they opposed it, and $\pm$ when they expressed ambivalence or division about the policy. Each lexicon operationalizes responsibility and science framing as detailed in Tables I and II.

C. Framing Detection Algorithm

The frame detection process uses a multi-stage algorithm aimed at enhancing both accuracy and computational efficiency. Initially, the text processing stage converts article headlines to lowercase and removes HTML tags using regular expressions. It preserves apostrophes to address contractions, eliminates non-alphanumeric characters while excluding spaces and hyphens, and normalizes whitespace. Subsequently, the pattern matching approach employs pre-compiled regular expressions to ensure that multi-word phrases such as 'operation warp speed' are matched as complete units, while word boundaries mitigate the risk of false matches, ensuring that "plan" will not match within "airplane".

Table II
SCIENCE FRAME LEXICONS

Positive Lexicon (50 terms)	Negative Lexicon (99 terms)
Expert Consultation & Deference	
followed cdc, cdc guidelines, followed guidelines, listened to experts, consulting experts, consulted scientists, expert advisors, scientific advisors, health advisors, sought expert advice, deferred to experts, heeded advice, followed recommendations, respected expertise	rejected science, rejected expert advice, ignored science, ignored experts, dismissed experts, dismisses science, goes against science, contradicts science, defies science, overruled scientists, disregards evidence, anti-science
Silencing & Suppression of Experts	
	fired experts, removed experts, sidelined scientists, muted scientists, gagged scientists, censored scientists, silenced experts, blocked scientists, stopped scientists, pushed out, marginalized experts, undermined scientists
Evidence-Based Decision Making	
evidence-based, science-based, data-driven, research-based, based on science, scientific approach, follows the science, guided by science, let science guide, scientific evidence, backed by research, supported by data	without evidence, no scientific basis, baseless, medically unfounded, scientifically unsound
Misinformation & False Claims	
	misinformation, spreads misinformation, disinformation, false information, conspiracy theory, conspiracy theories, promotes conspiracy, pseudoscience, unproven, debunked, false claims, misleading claims, unfounded claims
Expert Relationships	
dr fauci, anthony fauci, dr birx, deborah birx, cdc director, surgeon general, nih, niaid, national institutes, public health experts, epidemiologists, respects experts, trusts scientists, works with scientists	contradicted fauci, contradicts fauci, attacked fauci, attacks fauci, undermined fauci, undermines fauci, sidelined fauci, ignored fauci, dismissed fauci, contradicted cdc, contradicts cdc, undermined cdc, undermines cdc, overruled cdc, ignored cdc, against cdc advice, defied cdc, clashed with
Specific Treatments & Dangerous Claims	
	hydroxychloroquine, bleach, disinfectant, inject bleach, inject disinfectant, drink bleach, unproven treatments, unproven therapies, quack cures, snake oil, miracle cure, unapproved treatments, dangerous advice
Vaccine Development & Scientific Success	
operation warp speed, warp speed success, vaccine development, vaccine research, clinical trials, funded research, research funding, scientific breakthrough, public-private partnership, accelerated development	
Public Health Behaviors	
	anti-mask, refused mask, wont wear mask, mocked masks, ridiculed masks, mask skepticism, anti-vaccine, vaccine skepticism, vaccine hesitancy, mask resistance
Pandemic Denial & Minimization	
	hoax, democratic hoax, political hoax, witch hunt, fake news, media hoax, overblown, overhyped, exaggerated threat, exaggerated, just the flu, no worse than flu, will disappear, like magic, totally under control, nothing to worry about

The Net Framing Score (Equation 1) provides a normalized measure of framing valence:

$$NFS = \frac{P - N}{W} \quad (1)$$

$$Norm_{pos} = \frac{P}{W} \times 1000 \quad (2)$$

$$Norm_{neg} = \frac{N}{W} \times 1000 \quad (3)$$

$$Intensity = \frac{P + N}{W} \quad (4)$$

where P and N represent the counts of positive and negative lexicon terms respectively, and W denotes the total word count. The normalized positive and negative scores (Equations 2 and 3) are scaled per thousand words for interpretability.

The intensity metric (Equation 4) captures the overall strength of framing regardless of direction.

Classification thresholds determine framing direction, where positive framing is indicated by a Net Score greater than 0.001 (approximately 1 more positive than negative term per 1,000 words), negative framing is represented by a Net Score less than -0.001, neutral framing falls within the range of -0.001 to 0.001 (indicating balanced framing), and absent framing occurs when no frame terms are detected (with both positive and negative counts equal to zero).

For the Responsibility frame, positive values indicate credit attribution while negative values indicate blame attribution. For the Science frame, positive values indicate pro-science framing while negative values indicate anti-science framing. The normalized positive and negative scores (Equations 2

and 3) are scaled per thousand words for interpretability. The intensity metric (Equation 4) captures the overall strength of framing regardless of direction. This normalization approach enables fair comparisons across articles of varying lengths.

To ensure the reliability and validity of the lexicon-based approach, the analysis incorporates a robust statistical testing framework. For the statistical comparison framework, the analysis employs multiple tests: the independent samples t-test to compare mean net scores between The New York Times and The Guardian, complemented by the Mann-Whitney U test as a non-parametric alternative; Cohen's d effect size provides a standardized measure of difference magnitude. Furthermore, a Chi-square test of independence examines the differences in the distribution of framing directions (negative, positive, neutral, absent) across the outlets, while Cramér's V effect size measures the strength of association in contingency tables.

III. RESULTS

A. Overall Framing Distribution

In the analysis of media coverage surrounding COVID-19, several key observations emerge from Table III. Firstly, both The New York Times (NYT) and The Guardian exhibit a consistent pattern of negative framing, as illustrated by negative mean net scores across both frame categories. This suggests that when relevant language is employed, it more frequently assigns blame and highlights anti-science behaviors rather than crediting or promoting pro-science engagement. This indicates a tendency to focus on deficiencies rather than successes in the discourse about the pandemic.

Another significant finding is the high rate of frame absence across the articles, with frame absence reported between 84% and 95%. This trend indicates that most headlines and abstracts prioritize factual reporting and tend to shy away from explicit framing terminology. It appears that this absence is influenced by the concise nature of headlines and abstracts, with more elaborate framing occurring within the article bodies.

In terms of responsibility framing, The New York Times stands out with a higher average intensity, scoring 0.00646 compared to The Guardian's 0.00344. Furthermore, The New York Times reported a higher percentage of negative responsibility framing, with 11.7% of its articles characterized as negative, contrasting with The Guardian's 7.3%. These findings suggest that The New York Times is more engaged in assigning responsibility for pandemic-related issues with greater intensity.

Despite these differences, both outlets reflect remarkable convergence in their framing of scientific discourse. The mean net scores for the science frame are identical at -0.00101, and the direction distributions are similar, indicating a consensus in portraying Trump's relationship with scientific expertise during the pandemic.

Lastly, the analysis reveals an asymmetric relationship between credit and blame within responsibility framing. Both publications show a considerable inclination to assign blame over credit, with The New York Times exhibiting a blame-to-credit ratio of 5.3:1 and The Guardian at 3.3:1. Additionally,

anti-science language is more prevalent in both outlets, with ratios of 2.3:1 for The New York Times and 4.5:1 for The Guardian. These findings reinforce a significant bias toward criticism of actions and decisions rather than acknowledgment of positive efforts related to the pandemic response.

Table III
COMPARATIVE FRAMING ANALYSIS: NYT VS. GUARDIAN COVID-19 COVERAGE

Metric	NYT (n = 180)	Guardian (n = 8,627)
<i>Responsibility Frame</i>		
Mean Net Score	-0.0037	-0.0020
Standard Deviation	0.0147	0.0115
Median	0.0000	0.0000
Average Intensity	0.0065	0.0034
% Negative (Blame)	11.7	7.3
% Positive (Credit)	2.2	2.2
% Neutral	2.2	0.2
% Absent	83.9	90.2
<i>Science Frame</i>		
Mean Net Score	-0.0010	-0.0010
Standard Deviation	0.0087	0.0075
Median	0.0000	0.0000
Average Intensity	0.0022	0.0015
% Negative (Anti-Science)	5.0	3.7
% Positive (Pro-Science)	2.2	0.8
% Neutral	0.0	0.1
% Absent	92.8	95.4

B. Statistical Comparison Between Outlets

In Table IV, the results of the independent samples t-test reveal a statistically significant difference in responsibility framing between The New York Times and The Guardian ($t = -1.974$, $p = 0.0484$). The negative t-statistic indicates that The New York Times employs more blame-oriented framing than The Guardian, with a mean net score of -0.00372 compared to The Guardian's -0.00199.

Despite this significant difference, the Cohen's d effect size of -0.149 suggests that the practical significance is negligible. According to Cohen's (1988) classification, an effect size of $|d| < 0.2$ is considered negligible, while higher values indicate small, medium, or large effects. Thus, although statistically significant, the actual impact of this difference in framing is limited.

The Mann-Whitney U test, a non-parametric method that does not assume normal distributions, reports a p-value of 0.0520. This result is just above the conventional $\alpha = 0.05$ threshold, indicating a potential trend toward difference.

Additionally, a chi-square test examining the distribution of framing types (negative/positive/neutral/absent) shows a highly significant variation ($\chi^2 = 30.86$, $p < 0.0001$), signifying that the categorical distribution of framing differs systematically between the two outlets. The Cramér's V effect size of 0.059, while statistically significant, indicates only a small association.

Table IV
STATISTICAL COMPARISON OF FRAMING BETWEEN NYT AND GUARDIAN

Test	Responsibility Frame		Science Frame	
	Statistic	p-value	Statistic	p-value
Independent <i>t</i> -test	$t = -1.974$	0.0484*	$t = -0.013$	0.9897
Mann-Whitney <i>U</i>	$U = 742,787$	0.0520	$U = 776,442$	0.9992
Chi-square	$\chi^2 = 30.86$ ($df = 3$)	< 0.0001***	$\chi^2 = 4.95$ ($df = 3$)	0.1758
Cohen's <i>d</i>	-0.149		-0.001	
Cramér's <i>V</i>	0.059		0.024	
Interpretation	Significant difference		No significant difference	
Effect size	Negligible		Negligible	

Note: * $p < 0.05$, *** $p < 0.001$. Chi-square tests evaluate directional framing distribution.

The evidence suggests that The New York Times utilized more negative responsibility framing in its coverage of Trump's COVID-19 response than The Guardian, with a 60% higher likelihood of using blame-oriented language (11.7% vs. 7.3% of articles). While this finding is statistically significant and consistent across multiple tests, its practical significance is minimal. Both outlets largely employed negative responsibility framing when present, but The New York Times exhibited this tendency more frequently.

In contrast, the analysis of science framing shows no significant differences between the two outlets. The independent *t*-test yields $t = -0.013$ and $p = 0.9897$, indicating nearly perfect equivalence. Both media outlets share identical mean net scores of -0.00101, with a Cohen's *d* of -0.001 suggesting an insignificant effect.

The Mann-Whitney *U* test confirms this lack of significant difference ($p = 0.9992$), and the chi-square test assessing the distribution of framing directions is also non-significant ($p = 0.1758$), with a Cramér's *V* of 0.024 denoting negligible association.

The data strongly indicate convergent framing concerning Trump's relationship with scientific expertise across both outlets. The New York Times and The Guardian presented Trump's engagement with science quite similarly, showing comparable frequencies of anti-science language (5.0% vs. 3.7% of articles) and pro-science language (2.2% vs. 0.8%), along with overall negative net scores. This alignment suggests a shared perspective among elite media regarding Trump's pandemic response and its characterization as a rejection of scientific expertise.

C. Qualitative Analysis: Exemplar Articles

Table V presents key findings related to the framing of expectations surrounding the vaccine development program, particularly showcased by 'Operation Warp Speed.' This initiative serves as a notable case of sustained positive framing across both dimensions of responsibility credit and scientific achievement, consistently ranked among the most favorable examples.

Moreover, brief corrective articles tend to achieve high scores due to the concentrated use of specific terms (e.g.,

"death toll") without adequate contextual framing, underscoring a limitation in analyzing solely headlines and abstracts. The articles on related topics, such as the Democratic primaries and climate issues, employed a high density of specific terminology (e.g., "death tolls") while also considering their broader framing context, highlighting this same limitation.

Finally, articles featuring prominent figures such as Dr. Anthony Fauci, the CDC director, or other health officials frequently evoke positive scientific terminology, even in instances where the overall tone of the article may be critical of former President Trump. This observation suggests that the lexicon used reflects the presence of credible expertise rather than the article's sentiment.

D. Network Analysis: Frame Term Co-Occurrence Patterns

Beyond individual article analysis, a lexicon-based co-occurrence analysis was employed to examine how frame terms appear together within articles. Using Entman's (1993) framing theory, two specialized lexicons were created: a Responsibility Frame lexicon (50 positive and 88 negative terms) and a Science Frame lexicon (50 pro-science and 99 anti-science terms). These lexicons help analyze responsibility attribution for COVID outcomes and portrayals of credibility and expertise in science engagement. By applying regex-based pattern matching, terms from each lexicon were extracted from articles, forming co-occurrence networks that show which terms appeared together.

The bipartite network analysis identified 15 responsibility-science term pairs with co-occurrence frequencies of 2 or more. The strongest cross-frame connections centered on "operation warp speed," which co-occurred with science terms ($n=3$), demonstrating the integration of responsibility credit with scientific achievement narratives. Notably, blame-oriented terms showed consistent pairing with anti-science terminology: "failed" co-occurred with both "disinformation" and "misinformation" ($n=2$ each), while mortality terms ("toll," "death toll") co-occurred with "conspiracy theory," "bleach," and "disinfectant" ($n=2$ each). These patterns suggest that articles attributing blame for pandemic failures simultaneously emphasized anti-science behaviors such as promoting unproven treatments.

Table V
QUALITATIVE EXAMPLES OF FRAMING IN COVID-19 COVERAGE

Frame Category	Definition	Examples
Attribution of Responsibility (Most Negative)	This frame presents an issue or problem in such a way as to attribute responsibility for its cause or solution to either the government or to an individual or group.	<i>Guardian</i> : “Corrections and clarifications” - Blame terms: “toll”, “death toll” - Score: -0.286
Attribution of Responsibility (Most Positive)	This frame presents an issue or problem in such a way as to attribute credit for its solution to either the government or to an individual or group.	<i>Guardian</i> : “Trump unveils ‘warp-speed’ effort to create coronavirus vaccine by year’s end” - Credit terms: “operation warp speed”, “vaccine development”, “warp speed”, “rapid vaccine” - Score: $+0.111$
Science Frame (Most Negative)	Stories that emphasize skepticism toward scientific evidence, promote misinformation, or question expert consensus.	<i>Guardian</i> : “Twitter limits Donald Trump Jr’s account for posting Covid-19 misinformation” - Anti-science terms: “misinformation”, “hydroxychloroquine”, “false claims” - Score: -0.111
Science Frame (Most Positive)	Stories that emphasize trust in scientific evidence, expert guidance, and evidence-based approaches to managing the pandemic.	<i>Guardian</i> : “CDC director takes aim at Trump’s Covid adviser: ‘Everything he says is false’” - Pro-science terms: “anthony fauci”, “cdc director” - Score: $+0.069$

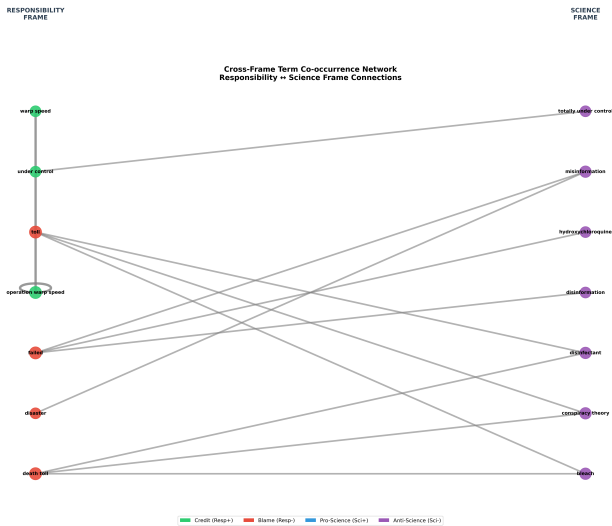


Figure 1. Cross Framing Network Graph

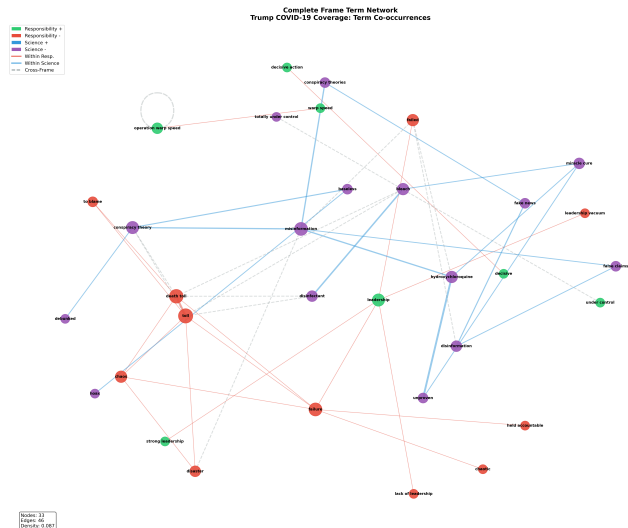


Figure 2. Full Frame Network Graph

The complete frame term network comprised 33 unique nodes and 46 edges (co-occurrences ≥ 2), with a network density of 0.087. Degree centrality analysis revealed the most structurally important terms: “toll” (0.250), “death toll” (0.219), “failure” (0.188), “misinformation” (0.188), and “leadership” (0.156). The prominence of mortality-related terms as network hubs indicates their central role in organizing both responsibility attribution and science-related discourse.

These central terms acted as narrative anchors, appearing across multiple framing contexts and linking otherwise

disparate coverage elements. The prominence of mortality-focused terms as network hubs indicates the central role that death statistics played in organizing both responsibility attribution and science-related discourse.

The relatively low network density (0.087) suggests selective rather than uniform co-occurrence patterns, indicating that journalists employed distinct framing packages for different aspects of pandemic coverage rather than randomly distributed terminology.

Comparative word cloud visualizations provided a direct visual comparison of frame term emphasis between The New York Times and The Guardian across all four categories under Responsibility +/-, Science +/-, revealing several notable patterns. Both outlets displayed a similar emphasis on mortality language, such as “death toll” and “toll,” as well as failure terminology in negative responsibility framing, indicating converging blame narratives. However, The Guardian showed greater prominence in accountability-focused terms like “held accountable” and “blame,” whereas NYT coverage highlighted process failures with terms like “no plan” and “delayed response.”

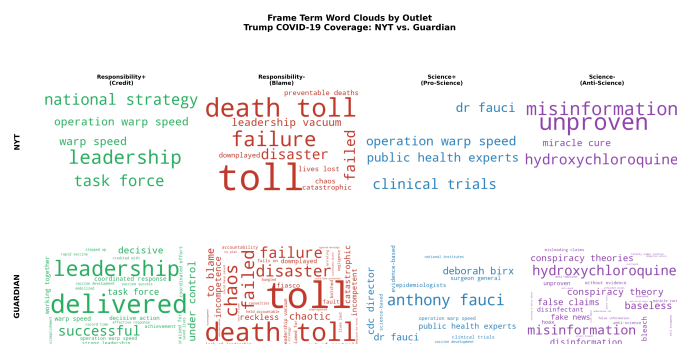


Figure 3. Word Cloud by Media Outlets

In terms of science framing, both outlets prominently featured “misinformation” and recognized expert figures, especially “Dr. Fauci.” However, The Guardian exhibited more diverse usage across multiple anti-science terms, while NYT focused on specific false claims, such as hydroxychloroquine promotion. Both outlets emphasized vaccine development and expert involvement in positive science framing, exhibiting relatively balanced term diversity. Additionally, the word clouds highlighted differences in terminology specificity, with The Guardian utilizing a more varied vocabulary across categories, while NYT’s tendency leaned toward the repeated use of fewer, more specific terms. This variation may reflect differing editorial approaches concerning frame consistency and lexical diversity.

IV. DISCUSSION

The analysis reveals statistically significant but practically small differences in responsibility attribution between The New York Times and The Guardian. The NY Times exhibited a 60% higher rate of blame-oriented articles (11.7% vs. 7.3%) and more negative mean net scores ($p = 0.0484$). However, the minimal Cohen’s d effect size (-0.149) indicates that while this difference is statistically measurable, its practical impact is modest. This finding partially contradicts expectations about the outlets’ left-leaning editorial stances. While one might anticipate uniform negative framing among progressive media, the NYT shows stronger blame attribution.

Several factors may explain this divergence. As the U.S. “newspaper of record,” [6] The New York Times may feel

a stronger obligation to hold the U.S. president accountable, while The Guardian, being UK-based, might adopt a more observational stance towards U.S. politics. The NYT’s smaller sample size ($n=180$) is concentrated around specific crisis periods, especially April to May 2020, potentially capturing moments of heightened accountability. In contrast, The Guardian’s larger and more evenly distributed sample ($n=8,627$) includes routine reporting, diluting average blame intensity.

In contrast to responsibility framing, the analysis shows nearly perfect convergence in science framing ($p = 0.99$, Cohen’s $d = -0.001$). Both outlets depicted Trump’s relationship with scientific expertise similarly, featuring comparable frequencies of anti-science language and identical mean net scores. This convergence indicates a unified global scientific assessment of Trump’s pandemic response, particularly regarding unproven treatments and conflicts with public health officials, which provided a shared foundation that transcended national media differences. Both outlets likely drew from overlapping scientific sources such as the WHO and CDC, facilitating consistent messaging about Trump’s anti-science behavior. Key events, such as Trump’s comments on disinfectants and public conflicts with Fauci, were so noticeable that they prompted similar framing across outlets.

Both outlets displayed significant negative framing. The NYT exhibited a 5.3:1 blame-to-credit ratio in responsibility framing and a 2.3:1 anti-science-to-pro-science ratio. The Guardian showed 3.3:1 and 4.5:1 ratios, respectively. This negativity bias can be attributed to several factors. Patterson’s research [7] highlights a media negativity bias in political reporting, influenced by structural elements that favor conflict and failure. This aligns with trends showing that news often highlights the unusual and contentious to capture audience attention. The blame-credit ratios also reflect substantive realities. The U.S. experienced high per-capita COVID-19 death rates, testing delays, and inconsistent public health messaging, leading to more accountability framing than success framing opportunities. The vaccine development program represents the primary positive framing, suggesting that while credit was given when achievements occurred, such opportunities were limited compared to failures.

Therefore, 84%-95% of articles showed frame absence, with no detectable frame terms in headlines and abstracts. This raises several implications. Headlines often prioritize factual information and attention-grabbing language over explicit evaluative framing. Many simply report events, such as “Trump holds briefing on pandemic,” without frame lexicon. API-provided abstracts focus on concise summaries emphasizing key facts, limiting the presence of frame terms. The high absence rate indicates that more extensive framing language likely appears in the article bodies rather than in the headlines or abstracts. Future research analyzing complete article texts may reveal significantly higher frame presence.

V. LIMITATIONS AND FUTURE RECOMMENDATIONS

The current study faces limitations including a notable sample size imbalance, with a ratio of 48:1 between The Guardian articles ($n=8,627$) and The New York Times articles ($n=180$). This disparity significantly constrains the statistical power necessary for robust cross-outlet comparisons. The limitations inherent in the NYT sample stem from API pagination restrictions, which allow only 500 requests per day and limit to 5 requests per minute. Consequently, this necessitates extended intervals of inactivity to gather additional articles. While both datasets are large enough to facilitate meaningful analysis, a more balanced collection would enhance the validity of comparative insights drawn from the respective articles.

Furthermore, a significant restriction of the analysis stems from the decision to include only headlines and abstracts, excluding substantial portions of the framing language present in the body of the articles. The 84%-95% frame absence rates observed likely reflect this methodological limitation rather than an actual absence of framing. A full-text analysis would therefore yield a more nuanced and comprehensive understanding of the framing strategies employed by each outlet. Moreover, the lexicon used in this study may not fully capture emergent framing language that arose post-development, potentially overlooking context-dependent meanings that could lead to miscoding. Temporal coverage is also restricted to the Trump administration, limiting insights into any shifts in framing that may have occurred post-Trump or during the Biden administration. Lastly, analyzing only two elite, left-leaning outlets restricts generalizability; insights derived from conservative platforms could reveal markedly different framing patterns, emphasizing the need for a broader examination of ideological diversity within media coverage.

The future directions for this study include several key approaches. First, a full-text analysis could be conducted by leveraging web scraping or full-text APIs, allowing for a more comprehensive assessment of framing across complete articles. This would likely uncover a significantly higher presence of frames and enable a more nuanced understanding of framing strategies. Additionally, extending the sleep time when using the New York Times API could facilitate the extraction of more articles. Second, broadening the outlet sample to incorporate ideologically diverse publications, such as conservative outlets like Fox News, the Wall Street Journal, and the New York Post, as well as international media from regions such as Asia, Latin America, and Africa, would enhance the ability to test hypotheses related to ideological and institutional influences on framing. Finally, extending the temporal coverage to include the years 2021-2024 would enable a comparative analysis of pandemic framing during the Trump and Biden administrations, assessing whether the identified patterns are specific to the Trump era or if they persist across different administrations.

VI. CONCLUSION

This study reveals that media framing of political accountability during crises shows significant variation influenced by

national context and institutional positioning. While responsibility framing diverges based on domestic and international perspectives, science credibility often converges, suggesting strong underlying consensus. The analysis highlights the critical role of diverse media perspectives in shaping public understanding, especially during acute crises. The findings emphasize the need for nuanced language in coverage to avoid exacerbating public fear or misinformation. Future research should explore broader contexts and methodologies to deepen insights into how media narratives impact perceptions of political leadership and accountability.

VII. ASSOCIATED CODE AND DATA

The code used for my analysis is contained in these files [8]: `src/frame_network_visualisation.py` and `src/two_dimension_framing_analysis.py`.

The following figures are included in the report:

- `results/word_clouds_by_outlet.png`
- `results/cross_frame_network.png`
- `results/full_frame_network.png`
- `results/framing_analysis_results.csv`
- `results/summary_statistics.csv`
- `results/temporal_analysis.csv`

[1]–[8]

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